Theory Assignment – 3

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Q1. What is the difference between frontend and backend JavaScript?

Frontend JavaScript runs in the user's browser and is used to make webpages interactive, such as validating forms or responding to button clicks. Backend JavaScript runs on a server, commonly using Node.js, and handles tasks such as processing requests, working with databases, and managing authentication.

Q2. What is DOM?

DOM stands for Document Object Model. It represents an HTML document as a tree of objects. JavaScript can use the DOM to access, change, add, or remove HTML elements and their styles.

Q3. What is an event in JavaScript?

An event is an action or occurrence that JavaScript can detect and respond to. Examples include clicking a button, typing on a keyboard, submitting a form, or loading a webpage.

Q4. What is the event loop in JavaScript?

The event loop helps JavaScript handle asynchronous tasks. When the call stack is free, it schedules completed asynchronous callbacks to run. This allows tasks such as timers and network requests to be handled without blocking the main thread while they are waiting.

Q5. What is hoisting in JavaScript?

Hoisting is JavaScript's behaviour of processing declarations before executing the code. Function declarations can be called before they appear in the code. Variables declared with var are initialized as undefined, while let and const remain unavailable before their declaration is reached.

Q6. What is event bubbling?

Event bubbling is the process in which an event starts at the target element and then moves upward through its parent elements. For example, clicking a button inside a div can trigger a click handler on the button and then on the div.

Q7. What is the difference between named, anonymous, and arrow functions?

A named function has a name and can be called using that name. An anonymous function has no function name and is often assigned to a variable or passed as a callback. An arrow function uses the => syntax and has a shorter form; it also uses the surrounding scope's this value.

```
// Named function
function greet() {
  return "Hello";
}

// Anonymous function
const sayHi = function() {
  return "Hi";
};

// Arrow function
const welcome = () => "Welcome";
```

Q8. Explain for-of and for-in loops.

The for-of loop is used to iterate over values in an iterable, such as an array or string. The for-in loop iterates over enumerable property keys of an object.

```
const fruits = ["Apple", "Mango", "Banana"];

for (const fruit of fruits) {
  console.log(fruit); // values
}

const student = { name: "Aman", age: 19 };

for (const key in student) {
  console.log(key); // property keys
}
```

Q9. Explain any five array methods.

1. push(): Adds one or more elements to the end of an array.

2. pop(): Removes the last element from an array.

3. shift(): Removes the first element from an array.

4. unshift(): Adds one or more elements to the beginning of an array.

5. map(): Creates a new array by applying a function to each element.

Q10. What is a callback method?

A callback is a function passed as an argument to another function so that it can be called later. Callbacks are commonly used for event handling and asynchronous operations.

```
function processNumber(number, callback) {
  callback(number);
}

processNumber(5, function(value) {
  console.log(value);
});
```